

CO3117 - W01 - W04 Baseline Diagnostic:

Foundations & Decision Tree

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W01 - W04

A. SPLIT, PIPELINE & Data Leakage

1> A valid pipeline

i) Raw UCI HAR data: 30 participants, each row is a window with a subject id.

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ii) Development pool: 24 participants in the standard split for example divide them into 16 for fitting and 8 for validation.

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iii) Fit & select: Fit preprocessing and the model on the fitting set
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select the configuration using validation.

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iv) Hold out official test: 8 separate participants; use for the final model evaluation after all choices are locked.

2> Why split by subject?

UCI HAR records accelerometer and gyroscope signals which are divided into 2.56 second windows with 50% overlap, so near-by windows can be highly correlated. If rows are split at random after windowing, samples from the same participant can end up in both train and test. The resulting score may not measure generalization to new people.

3. Ways to prevent leakage

1. Fit preprocessing on the fitting set only: fit the imputer, scaler, encoder, feature selector, and PCA using only the fitting portion
2. Reserve the test set from model decision
3. Split by group before training: same participant never appear on both sides.

B. Taxonomy and metrics

1. Distinguish the concepts pairs:
 - Supervised: learn from feature label pairs X and Y to predict label for new samples.
 - Unsupervised: uses only X to discover structure such as clusters, direction of variation.
 - Parametric: once the model form, dimensionality and structure are fixed, the number of parameter does not grow with the number of training samples.
 - Non-parametric: model complexity or structure can grow with the data
 - Generative: models joint distribution $P(X, y)$, often through $P(X|y)P(y)$
 - Discriminative: directly models $P(y|X)$ or the decision boundary
2. Classify the models
 - Decision Tree: supervised - non parametric - discriminative.
 - Naive Bayes: ——— - parametric - generative.

3. Accuracy & Macro-F1.

Accuracy = $\frac{\text{number of correct predictions}}{\text{total number of samples}}$

Macro-F1 =

If 90% of samples belong to the majority class, a model that always predict that class can achieve 90% accuracy while missing every other class.

→ Macro F1 computes F1 for each class and takes the arithmetic mean, giving every class equal weight.

C. Bias - Variance and Validation Curve.

1. Underfitting: the model is too simple to capture the data structure, so performance poor on both training and validation.

2. Overfitting: the model fits the training data too closely including noise; training performance is strong but validation performance is noticeably worse.

Decision tree

Impurity and Splits: ^{+ class proportions p_k} Entropy and Gini formula for a set S , C class

1. Formulas: $H(S) =$

$$G(S) =$$

2. The parent node has 8 samples: 6 class A and 2 class B
After the split, the left child has 4A/0B and the right child has 2A/2B

Parent node: 6A, 2B

$$p_A = 6/8 = 0,75 ; p_B = 2/8 = 0,25$$

$$H(\text{parent}) = -0,75 \log_2 0,75 - 0,25 \log_2 0,25 = 0,811$$

$$G(\text{parent}) = 1 - (0,75^2 + 0,25^2) = 0,375.$$

Left child 4A, 0B.

$$p_A = 1 ; p_B = 0$$

$$H(\text{left}) = 0 \text{ bits} \Rightarrow G(\text{left}) = 0$$

the node is pure class A.

Right child 2A, 2B

$$p_A = p_B = 0,5 . H(\text{right}) = 1 \text{ bit} ; G(\text{right}) = 1 - (0,5^2 + 0,5^2) = 0,5.$$

Weighted child impurity

$$w_L = w_R = 4/8 = 0,5$$

$$H(\text{children}) = 0,5 \times 0 + 0,5 \times 1 = 0,5 \text{ bits}$$

$$G(\text{children}) = 0,5 \times 0 + 0,5 \times 0,5 = 0,25.$$

$$\text{Information gain} = 0,811 - 0,5 = 0,311 \text{ bits}$$

$$\text{Gini reduction} = 0,375 - 0,25 = 0,125.$$

A. Interpretation

The split reduce weight impurity ; Information gain is about 0,311 bits and Gini decreases by 0,125

The left child is pure class A ; the right child remains 50/50 , giving that child the maximum impurity

for a two class problem. These are weighted average reductions across the children; ~~they do not~~

E. Continuous features, Missing Values and Pruning.

1. Splitting continuous features.

The tree tests a threshold t and splits samples into two groups: $x_j \leq t$ and $x_j > t$. Given the sorted distinct feature values $x^{(1)} < \dots < x^{(n)}$, candidate thresholds are commonly the midpoints between adjacent values.

$$t_i = (x^{(i)} + x^{(i+1)}) / 2$$

2. Missing values and leakage.

~~Let HAR~~ For a dataset that does ~~not~~ have missing values one option is to impute numeric values with the median and categorical values with the mode.

3. Pre-pruning and post pruning.

- pre-pruning / early stopping: Stop the tree from growing when it reaches limit such as max-depth, min-sample or a minimum impurity improvement. It tends to increase bias and reduce variance; Stopping too early can miss useful structure.

- post-pruning: lets the tree grow first, then removes less useful branches using a validation criterion or cost-complexity pruning. It tends to reduce complexity and variance; also raises bias.

4. Why use cross entropy (NLL) but not MSE.

* Decision Tree

Classification tree: leaf store a class label or a class distribution.

Regression Tree: leaf store a numeric value.

Entropy: $H(S) = - \sum_{i=1}^K p_i \log_2 p_i$ / p_i : proportion in class

(measure the uncertainty or impurity of a dataset S)

Information gain: $IG(S, A) = H(S) - \sum_{v \in \text{values}(A)} \frac{|S_v|}{|S|} H(S_v)$

S : dataset; A : attribute used to split. S_v : subset of S where attribute $A = v$.

Measure the reduction in entropy after splitting by A .

Gini Index: $Gini(S) = 1 - \sum_{i=1}^K p_i^2$

. Another measure of impurity used in CART.

. $Gini(S) = 0 \rightarrow$ one class (pure)

ID3: use Entropy and IG

. Greedy approach: choose the attribute with the highest IG

. Recursive splitting until: All samples in node belong to the same class, or no attribute remain.

Limitations: - overfit when grow deep.

- bias Tôn trọng - Hợp tác - Khoan dung



Ch. 5:

Gain Ratio: reduce overfitting

$$GR(S, A) = \frac{IG(S, A)}{Split\ Info(S, A)}$$

$$Split\ Info(S, A) = - \sum_{v \in Values(A)} \frac{|S_v|}{|S|} \log_2 \frac{|S_v|}{|S|}$$

Step 1: Determine candidate threshold

— 2: Split dataset.

— 3: Calculate IG

— 4: Recursive 1 → 3

— 5: Choose threshold T that is optimal (highest IG)

pre pruning: $|S| < N_{min}$

post pruning: bottom up

$$\text{pessimistic error rate: } \hat{p}_{upper} \approx \hat{p} + z \cdot \sqrt{\frac{\hat{p}(1-\hat{p})}{N}}$$